Lab 4: dplyr, Part 1

Filter, slice, arrange, select, mutate, and summarize flights.

Work in a **Quarto** document. Number the exercises. Do **4.1** through **4.3** first.

```
library(tidyverse)
library(nycflights13)
data("flights")
```

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1. Filter and slice

1. Flights in odd months, or on even days of even months.
2. Flights to Houston (`IAH` or `HOU`), on UA, AA, or DL, in July–September, that arrived more than 15 minutes late and did not leave late.
3. Rows 10, 100, 1000, 10000, and 100000.
4. The last 10 of the first 30 Newark (`EWR`) flights.
5. `set.seed(1)`. A random 1% sample, and a sample of 3367 rows. Are the last 10 rows the same? Why?

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2. Arrange, select, mutate

1. Longest `distance`, ties broken by `air_time`.
2. Keep every arrival column plus `year`, `month`, and `day`.
3. Convert `dep_time` and `sched_dep_time` to minutes past midnight.
4. Move columns ending in `"time"` to just before `flight`. Then put `sched` columns before the actual times.

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3. Summarize and group

1. Standard deviation of `dep_delay`.
2. Maximum `dep_delay` by `origin` in one pipeline.
3. For each `day`, the number and proportion of canceled flights (`is.na(dep_delay) | is.na(arr_delay)`). Plot the proportion against day, and against average departure delay.